

Assignment 4: String Processing

1 Introduction

String matching algorithms are essential techniques used to find occurrences of a pattern inside a given text. They are widely used in search engines, DNA sequence analysis, plagiarism detection, and data processing systems.

In this assignment, we implement and analyze two advanced string matching algorithms:

- **Boyer-Moore Algorithm**
- **Rabin-Karp Algorithm**

We also compare their performance based on execution time, number of comparisons, and efficiency.

2 Selected Algorithms

2.1 Boyer-Moore Algorithm

Idea:

The Boyer-Moore algorithm compares the pattern from right to left and skips unnecessary comparisons using the bad character rule.

Key Features:

- Efficient for large texts
- Skips sections of text instead of checking every character
- Performs fewer comparisons in practice

2.2 Rabin-Karp Algorithm

Idea:

Rabin-Karp uses hashing to compare the pattern with substrings of the text. If hash values match, a character-by-character check is performed.

Key Features:

- Efficient for multiple pattern searching
 - Uses rolling hash technique
 - Good average performance
-

3 Complexity Analysis

Algorithm	Time Complexity	Space Complexity
Boyer-Moore	$O(n/m)$ best case	$O(1)$
Rabin-Karp	$O(n + m)$ average case	$O(1)$

Where:

- n = length of text
- m = length of pattern

4 Input File Specification

File Name: `input.txt`

Example Content:

```
DATA STRUCTURE AND ALGORITHMS COURSE
THIS COURSE TEACHES DATA HANDLING AND TREE STRUCTURES
RABIN KARP AND BOYER MOORE ARE STRING MATCHING ALGORITHMS
DATA DATA DATA STRUCTURE STRUCTURE
AAAAAA TEST FOR OVERLAPPING AAA PATTERN
```

5 Required Program Features

5.1 Interactive Menu

The program must display:

1. Load Text File
2. Enter Text Manually
3. Search Using Boyer-Moore
4. Search Using Rabin-Karp
5. Compare Algorithms
6. Exit

5.2 Multiple Pattern Search

The system should allow searching for:

- Single pattern
- Multiple patterns in one run

5.3 Pattern Highlighting

When a match is found, the pattern must be highlighted:

Example:

Input: DATA STRUCTURE COURSE

Output: DATA [STRUCTURE] COURSE

5.4 Overlapping Matches Support

The program must detect overlapping patterns correctly.

Example:

Text: AAAAAA

Pattern: AAA

Output:

Match at index: 0

Match at index: 1

Match at index: 2

5.5 File Processing

- Load text from external file
- Allow user to enter file name
- Search inside file content

5.6 Statistics Output

After each search, display:

- Number of comparisons
 - Number of matches
 - Execution time (ms)
 - Algorithm used
-

6 Implementation Details

Required Functions:

```
void loadFile();  
void manualInput();  
void boyerMooreSearch();  
void rabinKarpSearch();  
void compareAlgorithms();  
void printStatistics();
```

7 Algorithm Implementations (Summary)

7.1 Boyer-Moore Algorithm

- Starts comparison from the end of the pattern
- Uses bad character rule for shifting
- Skips unnecessary comparisons

7.2 Rabin-Karp Algorithm

- Computes hash of pattern and text substring
 - Compares hashes first
 - Verifies character match if hash matches
-

8 Test Cases

Text	Pattern	Expected Result
AAAAAAAAAA	AAA	Multiple overlapping matches
DATA STRUCTURE	STRUCTURE	Match found
HELLO WORLD	TEST	No match
MISSISSIPPI	ISSI	Multiple matches
ALGORITHM	RITH	No match
